

California Polytechnic State University, San Luis Obispo
Orfalea College of Business

NBA Player Salary

An Understanding and Analysis of NBA players' Salaries



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Abstract

Player salaries in the NBA are influenced by a wide range of factors, including individual performance, experience, position, team context, and prior contract history. Understanding how these variables interact can provide insight into how teams assign market value to players and how performance translates into compensation. This has important applications in sports analytics, contract evaluation, roster construction, and salary forecasting. The goal of this project is to develop machine learning models that predict an NBA player's salary in $t+1$ using information available through season t . To do this, we use a large dataset containing player statistics, advanced metrics, team-level indicators, service and experience information, and archetype-based features. We compare multiple supervised learning models including XGBoost, CatBoost, LightGBM, and blended ensemble models, across both overall and settings that are specific to a certain type of position. In addition, we look into two modeling strategies: one that includes salary-history variables for maximum predictive accuracy, and another performance-based model that excludes prior salary information in order to better isolate the relationship between on-court performance and future earnings. By evaluating these models on recent holdout seasons, this project aims to identify the strongest predictors of salary and demonstrate how machine learning can be used to estimate player value in a more systematic and interpretable way.



Type of Project

This project is a multivariable regression problem. We use a variety of machine learning models and player performance data to predict an NBA player’s salary in the upcoming season. Because salary is a continuous numeric outcome, the project is a predictive regression problem, with additional comparisons across model types and player positions.

Data

Data Source:

The dataset for this project is a fully cleaned and merged NBA player-season panel dataset (one row per player per season, 2000–2025; ~8,336 rows) built from Basketball-Reference and Spotrac sources amongst others. It combines on-court performance metrics, advanced impact statistics (e.g., PER, BPM, VORP, WS), availability/durability measures (including games missed and multi-year trends), role/context features (primary position group and archetype indicators), and experience/CBA-bucket variables. A central challenge is balancing predictive power with interpretability while handling mixed numeric/categorical features and strong season-to-season nonlinearity in salary outcomes. To reduce noise and leakage, features were engineered to be time-safe (lagged and rolling history only, no predicting on current season), categorical fields were standardized into model-ready groupings/flags, and missingness was minimized through consistent cleaning rules. The modeling target is next-season salary, and data splitting is done chronologically so only past seasons are used to train/tune models before evaluation on future seasons.

Data Description

Variable Descriptions

Table 1. Variable Names and Descriptions

<u>Variable Name</u>	<u>Description</u>
<u>adv_ast_to</u>	<u>Advanced assist-to-turnover metric.</u>
<u>adv_def_rating</u>	<u>Advanced defensive rating (lower is better).</u>
<u>adv_net_rating</u>	<u>Advanced net rating (offensive minus defensive impact).</u>
<u>adv_true_shooting_pct</u>	<u>True shooting percentage (2PT, 3PT, FT efficiency combined).</u>
<u>adv_usage_pct</u>	<u>Usage rate (% of team possessions used while on court).</u>
<u>age</u>	<u>Player age in that season.</u>
<u>all_star_selections_through_prev_season</u>	<u>Cumulative All-Star selections up to the prior season.</u>
<u>archetype_confidence_score</u>	<u>Confidence score for assigned archetype label.</u>
<u>archetype_is_balanced_connector</u>	<u>1 if player archetype is balanced connector.</u>
<u>archetype_is_interior_big_finisher</u>	<u>1 if player archetype is interior big finisher.</u>
<u>archetype_is_lead_creator_guard</u>	<u>1 if player archetype is lead creator guard.</u>

<u>archetype_is_rim_protector_big</u>	<u>1 if player archetype is rim protector big.</u>
<u>archetype_is_scoring_guard_wing</u>	<u>1 if player archetype is scoring guard/wing.</u>
<u>archetype_is_stretch_big</u>	<u>1 if player archetype is stretch big.</u>
<u>archetype_is_three_and_d_wing</u>	<u>1 if player archetype is three-and-D wing.</u>
<u>archetype_scarcity_index</u>	<u>Season-level scarcity score for that archetype.</u>
<u>archetype_share_in_season</u>	<u>Share of players in that archetype during the season.</u>
<u>assists_pg</u>	<u>Assists per game.</u>
<u>bbr_bpm</u>	<u>Basketball-Reference Box Plus/Minus.</u>
<u>bbr_per</u>	<u>Basketball-Reference Player Efficiency Rating.</u>
<u>bbr_vorp</u>	<u>Basketball-Reference Value Over Replacement Player.</u>
<u>bbr_ws_per_48</u>	<u>Basketball-Reference Win Shares per 48 minutes.</u>
<u>blocks_pg</u>	<u>Blocks per game.</u>

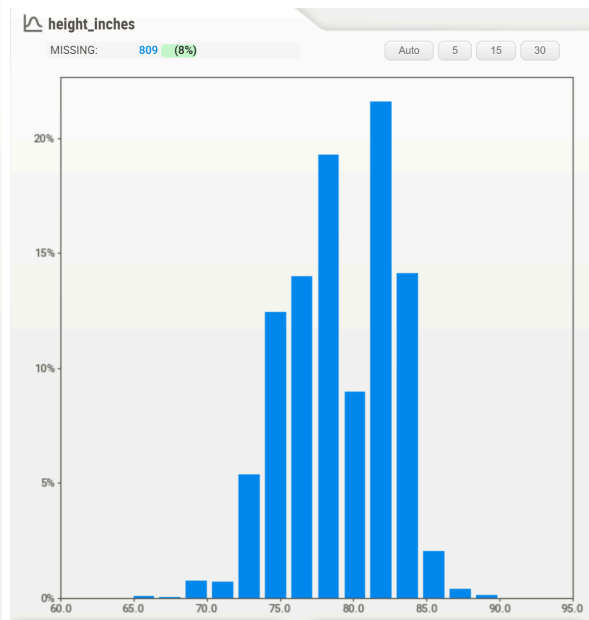
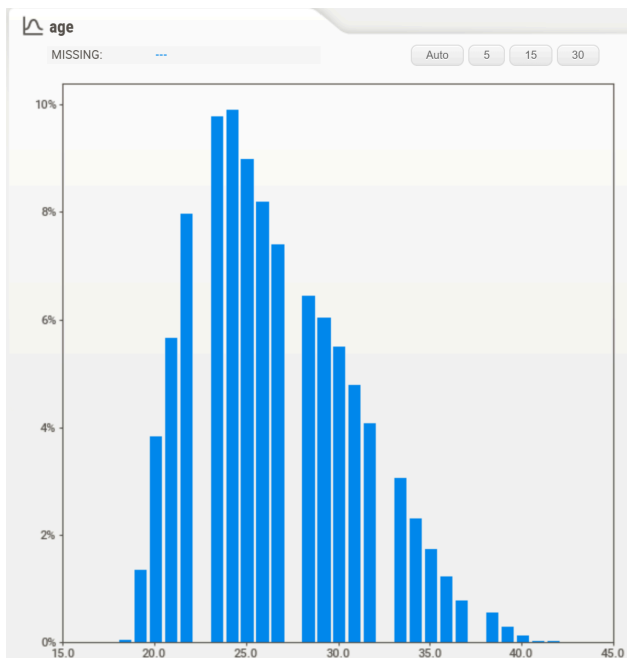
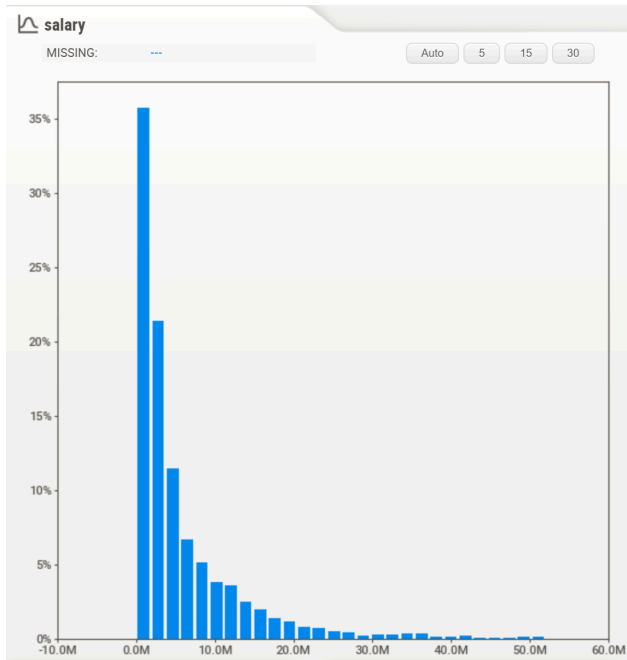
<u>cba_bucket_is_0_2</u>	<u>1 if years-of-service bucket is 0-2.</u>
<u>cba_bucket_is_10_plus</u>	<u>1 if years-of-service bucket is 10+.</u>
<u>cba_bucket_is_3_6</u>	<u>1 if years-of-service bucket is 3-6.</u>
<u>cba_bucket_is_7_9</u>	<u>1 if years-of-service bucket is 7-9.</u>
<u>def_rebounds_pg</u>	<u>Defensive rebounds per game.</u>
<u>effective_fg_pct</u>	<u>Effective field-goal percentage.</u>
<u>fg_attempted_pg</u>	<u>Field-goal attempts per game.</u>
<u>fouls_pg</u>	<u>Personal fouls per game.</u>
<u>ft_attempted_pg</u>	<u>Free-throw attempts per game.</u>
<u>ft_pct</u>	<u>Free-throw percentage.</u>
<u>games</u>	<u>Games played.</u>
<u>games_missed</u>	<u>Estimated games missed in the season.</u>
<u>games_missed_rate</u>	<u>Games missed as a proportion of team games.</u>
<u>games_started</u>	<u>Games started.</u>

<u>height_inches</u>	<u>Height in inches.</u>
<u>minutes_per_game</u>	<u>Minutes per game.</u>
<u>off_rebounds_pg</u>	<u>Offensive rebounds per game.</u>
<u>points_pg</u>	<u>Points per game.</u>
<u>position_is_combo</u>	<u>1 if player has combo/multi-position designation.</u>
<u>prev_all_def_any</u>	<u>Prior-season indicator for All-Defense selection.</u>
<u>prev_all_nba_any</u>	<u>Prior-season indicator for All-NBA selection.</u>
<u>prev_assists_pg</u>	<u>Prior-season assists per game.</u>
<u>prev_award_all_star</u>	<u>Prior-season All-Star indicator.</u>
<u>prev_minutes_per_game</u>	<u>Prior-season minutes per game.</u>
<u>prev_points_pg</u>	<u>Prior-season points per game.</u>
<u>prev_rebounds_pg</u>	<u>Prior-season rebounds per game.</u>
<u>prev_team_win_pct_regular</u>	<u>Prior-season team regular-season win percentage.</u>

<u>rebounds_pg</u>	<u>Rebounds per game.</u>
<u>reg_plus_minus_pg</u>	<u>Plus/minus per game (regular season).</u>
<u>rolling3_points_pg_prior</u>	<u>Trailing 3-season average of points per game prior to current season.</u>
<u>season</u>	<u>Season-end year for the current row.</u>
<u>steals_pg</u>	<u>Steals per game.</u>
<u>team_net_points_pg</u>	<u>Team net points per game.</u>
<u>team_region_central</u>	<u>1 if team is mapped to central region.</u>
<u>team_region_east</u>	<u>1 if team is mapped to east region.</u>
<u>team_region_west</u>	<u>1 if team is mapped to west region.</u>
<u>team_win_pct_regular</u>	<u>Team regular-season win percentage.</u>
<u>three_pt_attempted_pg</u>	<u>Three-point attempts per game.</u>
<u>three_pt_pct</u>	<u>Three-point percentage.</u>
<u>turnovers_pg</u>	<u>Turnovers per game.</u>

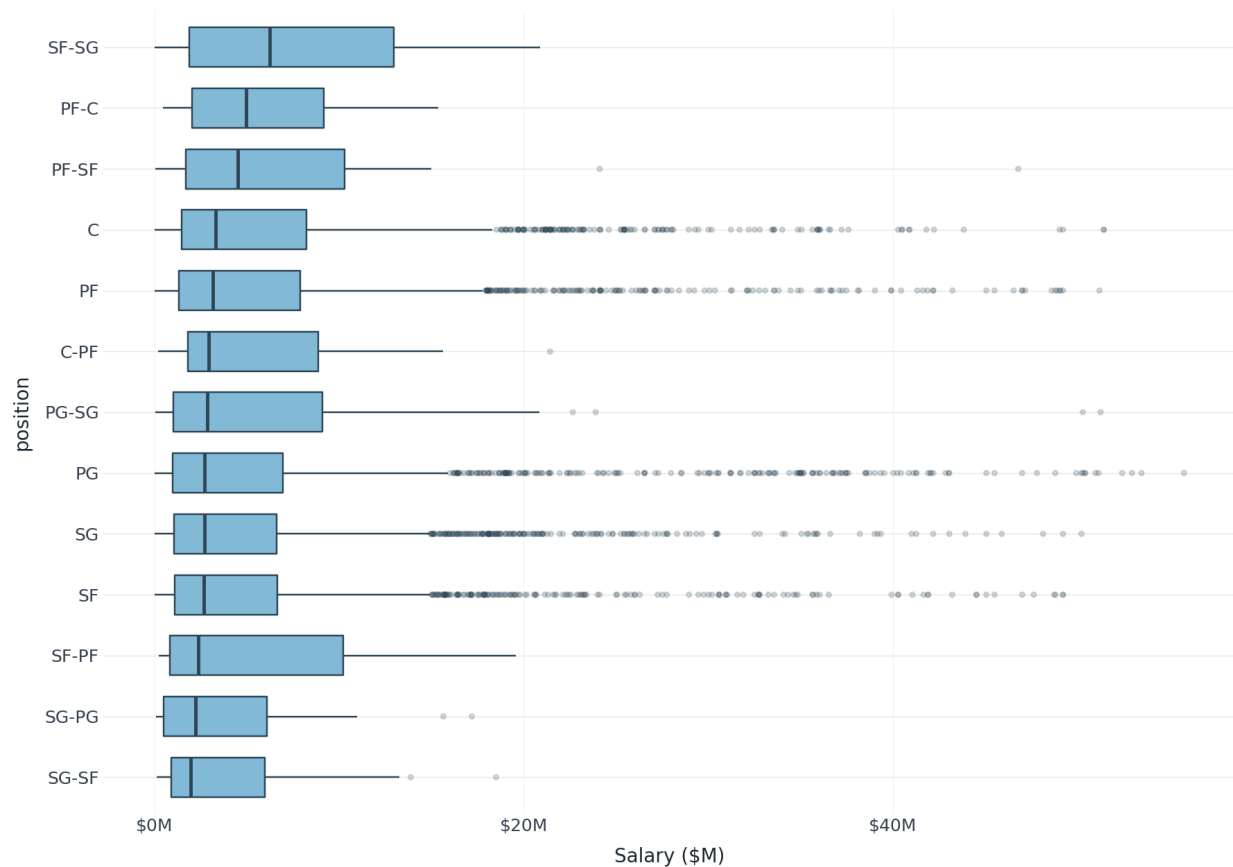
<u>weight_lbs</u>	<u>Weight in pounds.</u>
<u>years_of_service</u>	<u>NBA years of service for CBA/experience context.</u>

Data Visualizations



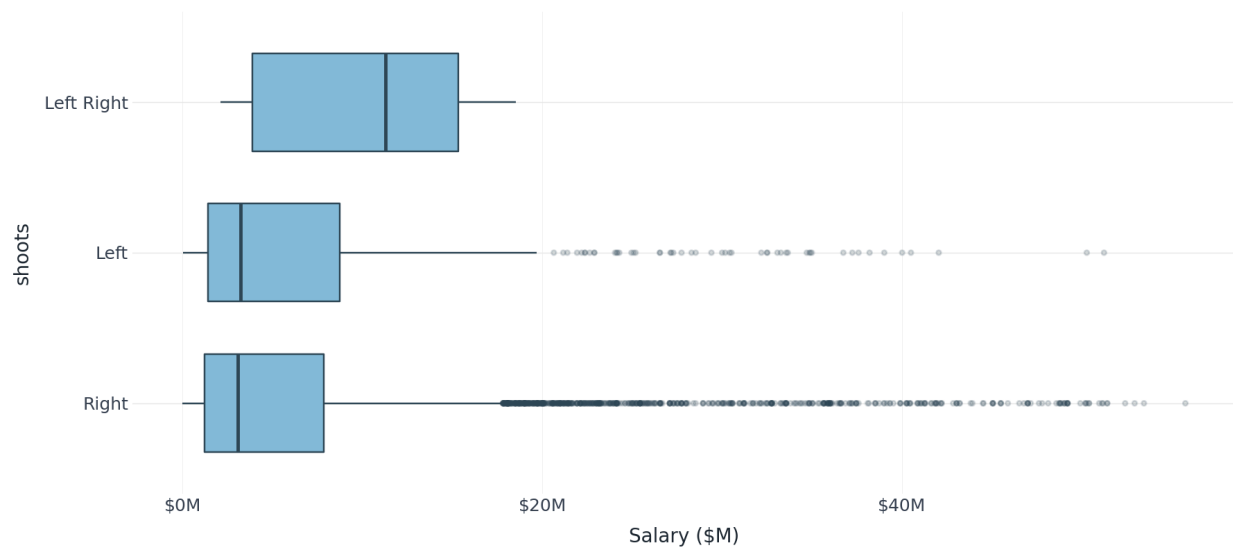
NBA Salary by Position

Categories ordered by median salary; groups with fewer than 10 rows are excluded



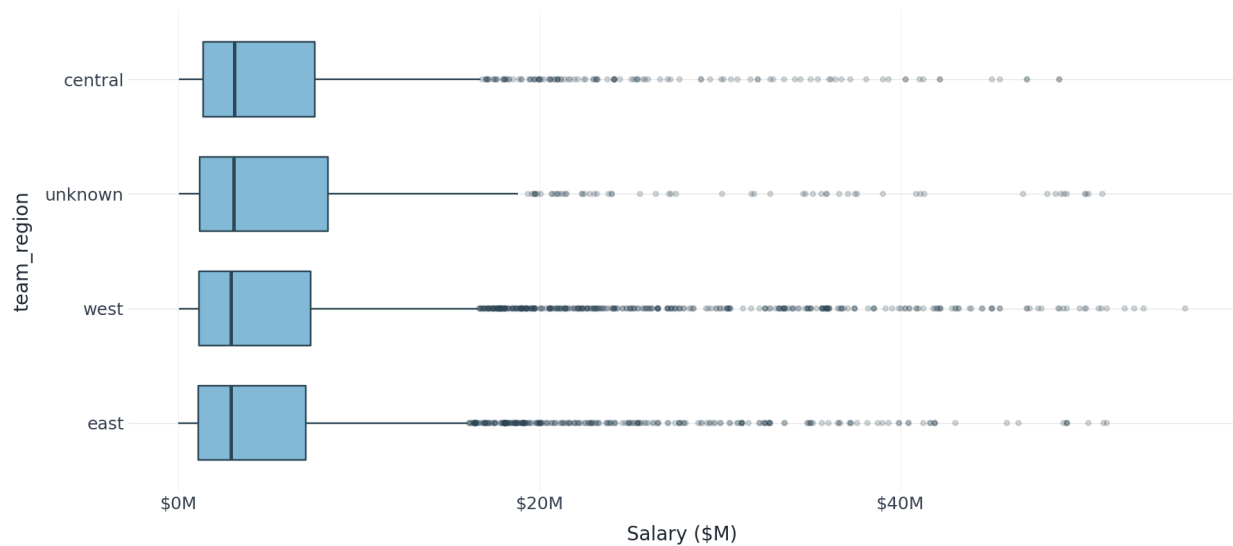
NBA Salary by Shoots

Categories ordered by median salary; groups with fewer than 10 rows are excluded



NBA Salary by Team Region

Categories ordered by median salary; groups with fewer than 10 rows are excluded



NBA Salary by Contract Type

Categories ordered by median salary; groups with fewer than 10 rows are excluded



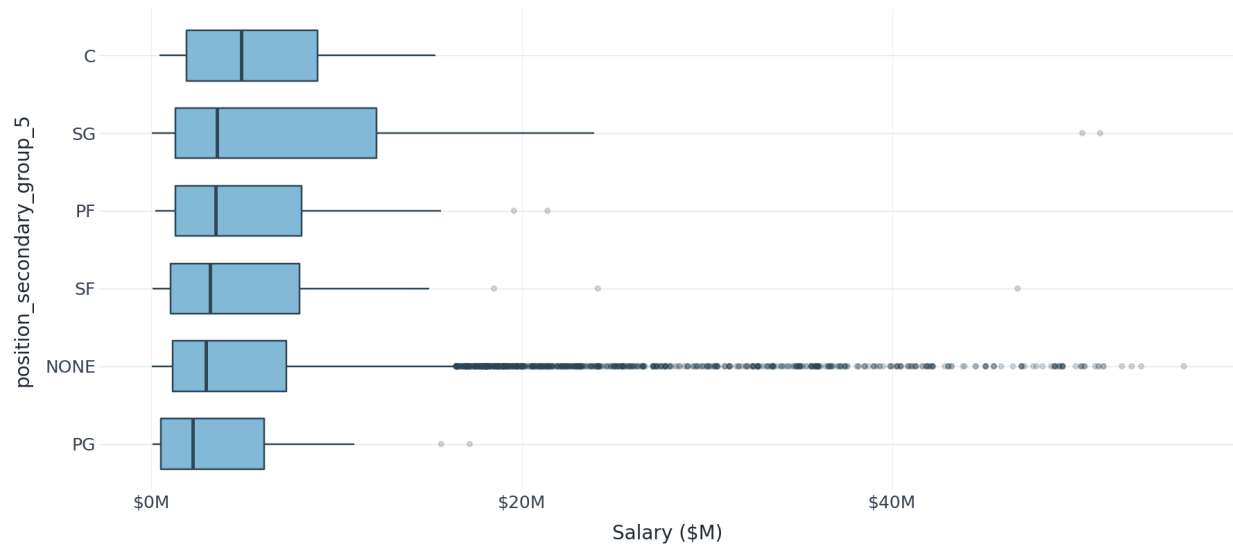
NBA Salary by Position Primary Group 5

Categories ordered by median salary; groups with fewer than 10 rows are excluded



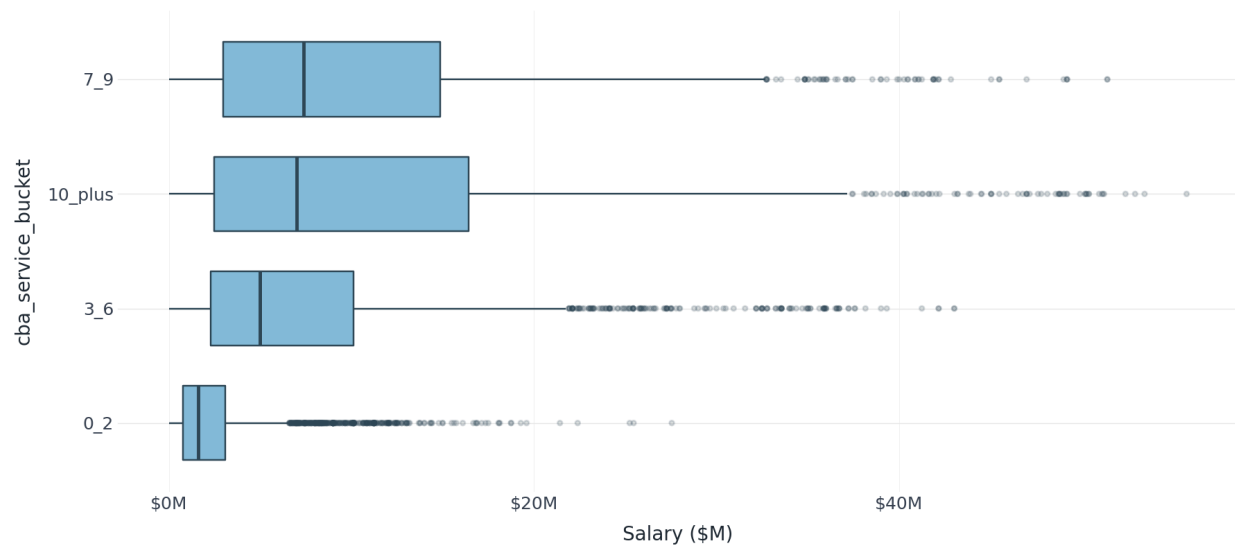
NBA Salary by Position Secondary Group 5

Categories ordered by median salary; groups with fewer than 10 rows are excluded



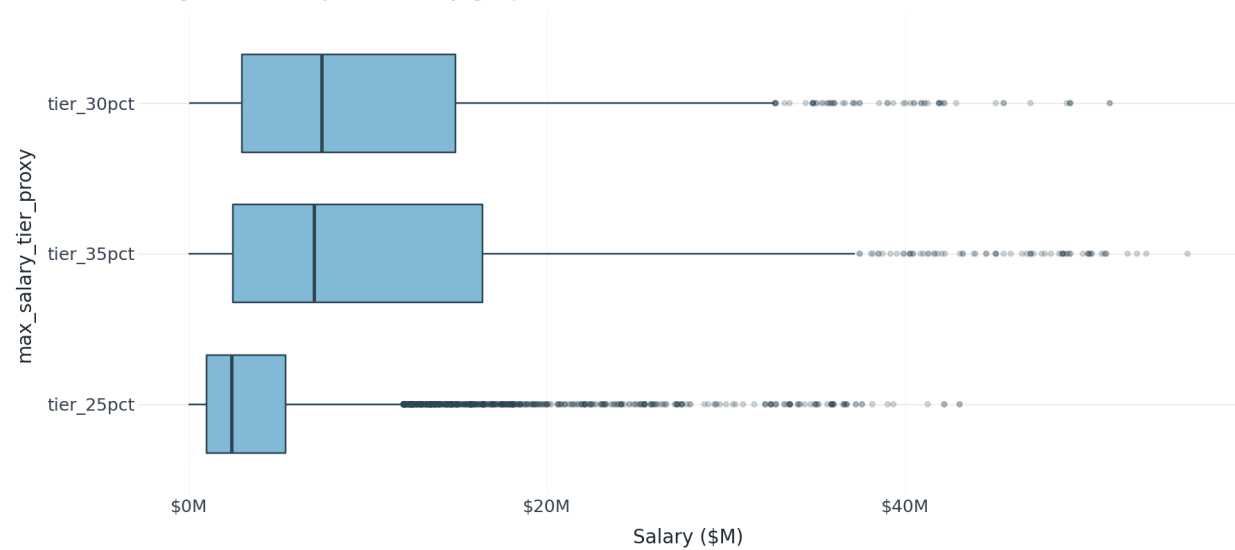
NBA Salary by Cba Service Bucket

Categories ordered by median salary; groups with fewer than 10 rows are excluded



NBA Salary by Max Salary Tier Proxy

Categories ordered by median salary; groups with fewer than 10 rows are excluded



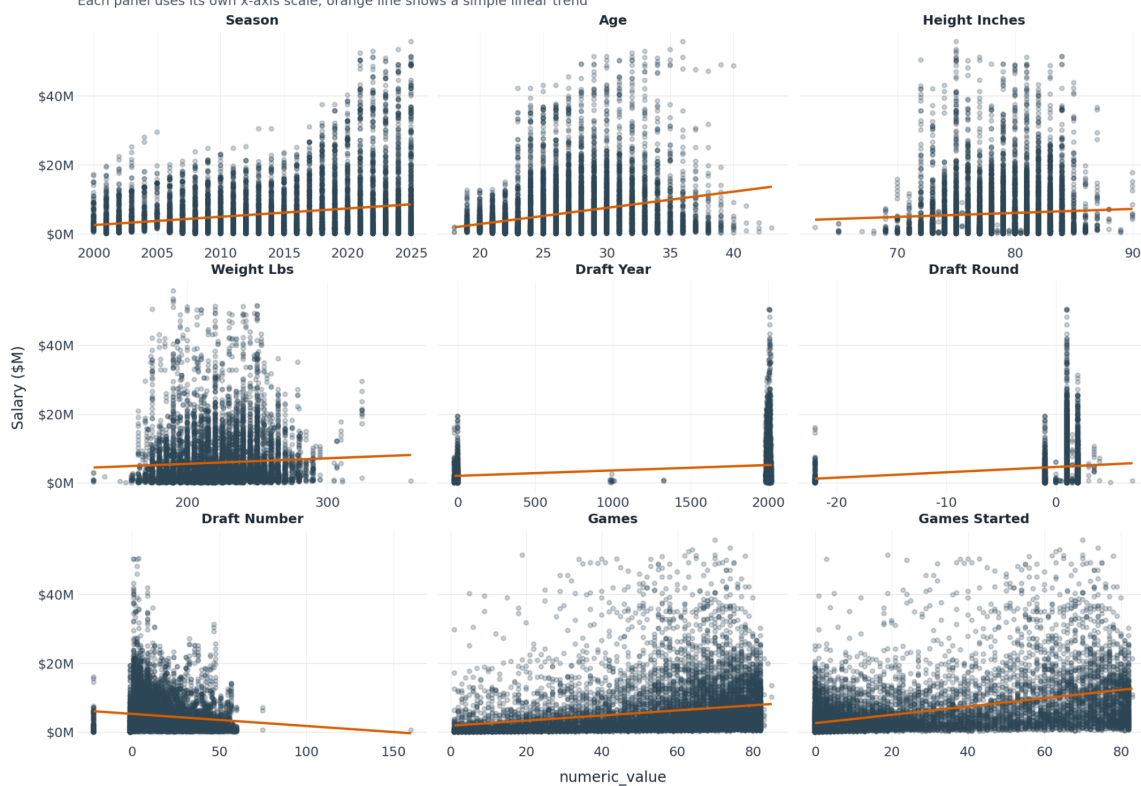
NBA Salary by Archetype Label

Categories ordered by median salary; groups with fewer than 10 rows are excluded



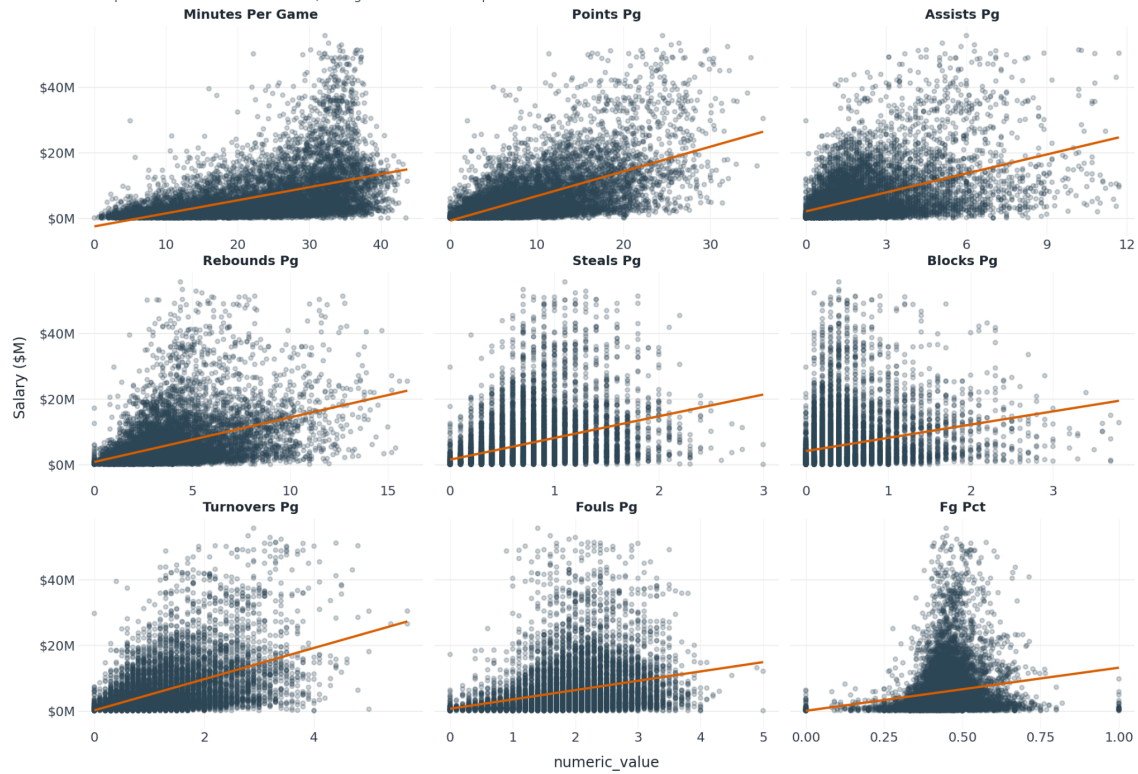
NBA Salary vs Numeric Variables (1 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



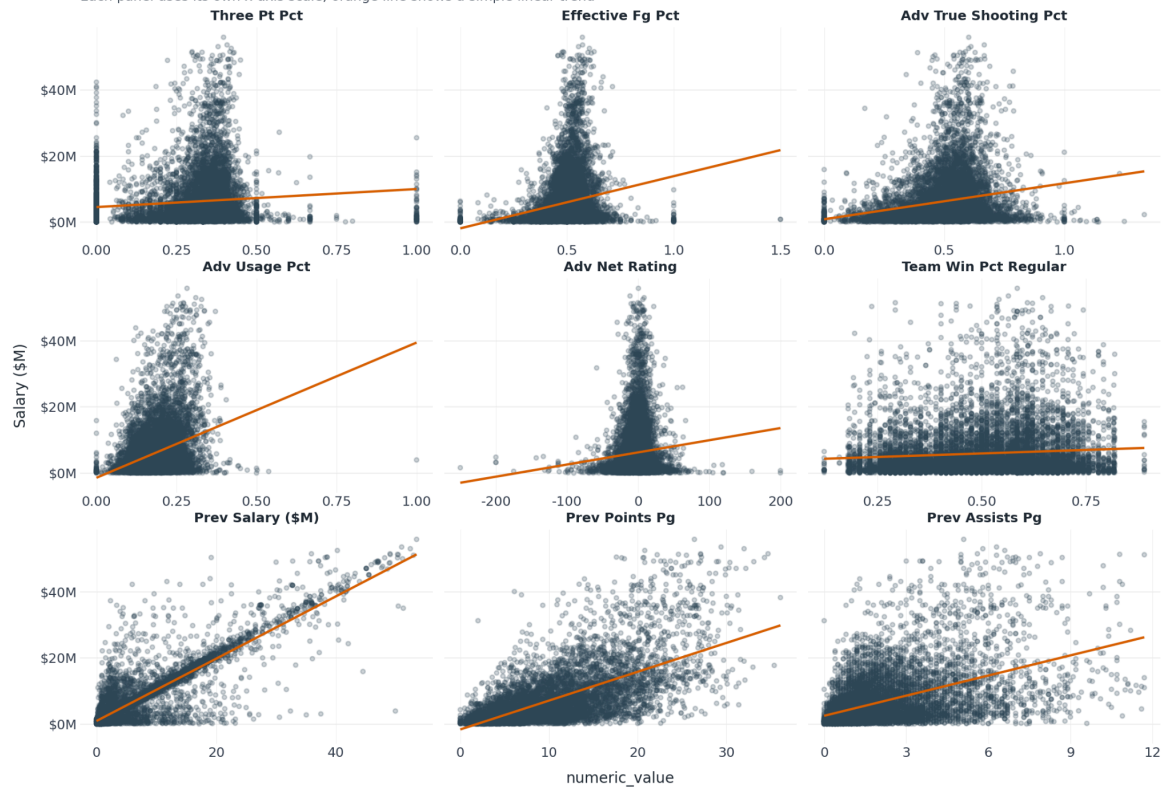
NBA Salary vs Numeric Variables (2 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



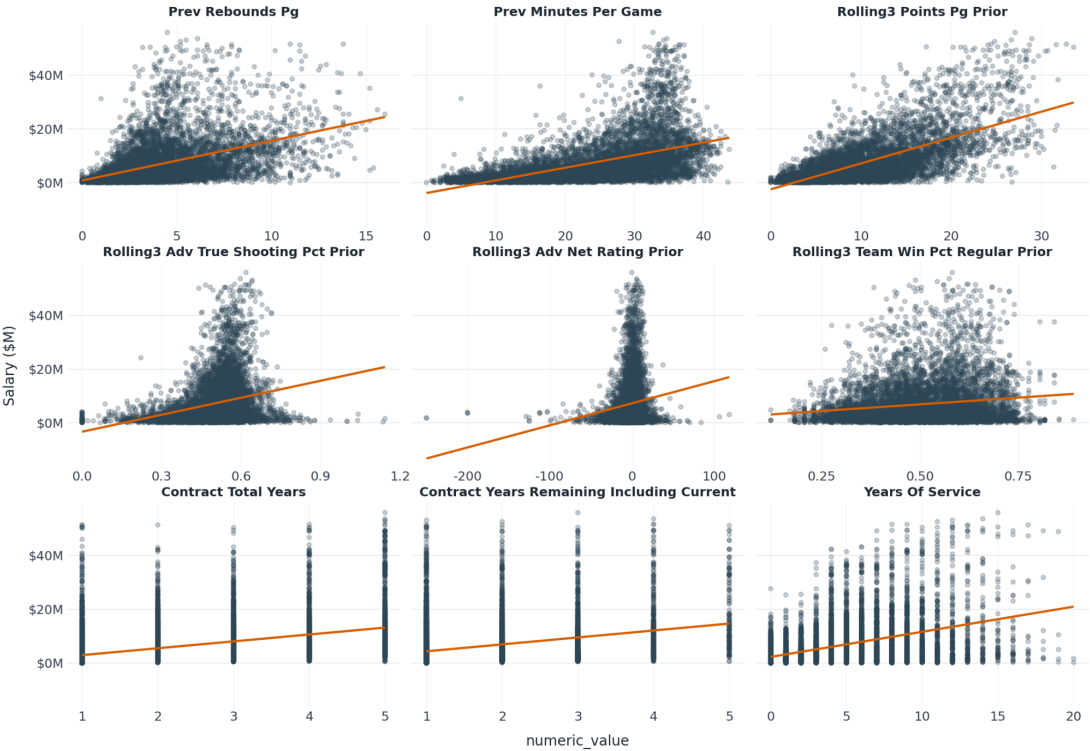
NBA Salary vs Numeric Variables (3 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



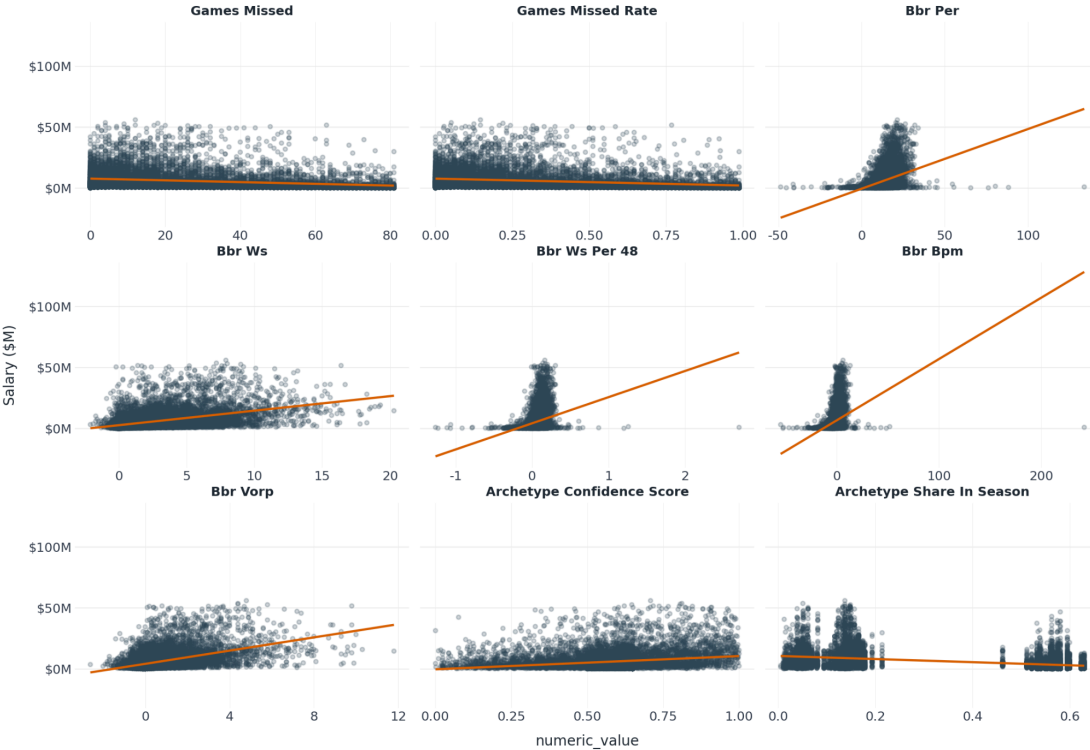
NBA Salary vs Numeric Variables (4 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



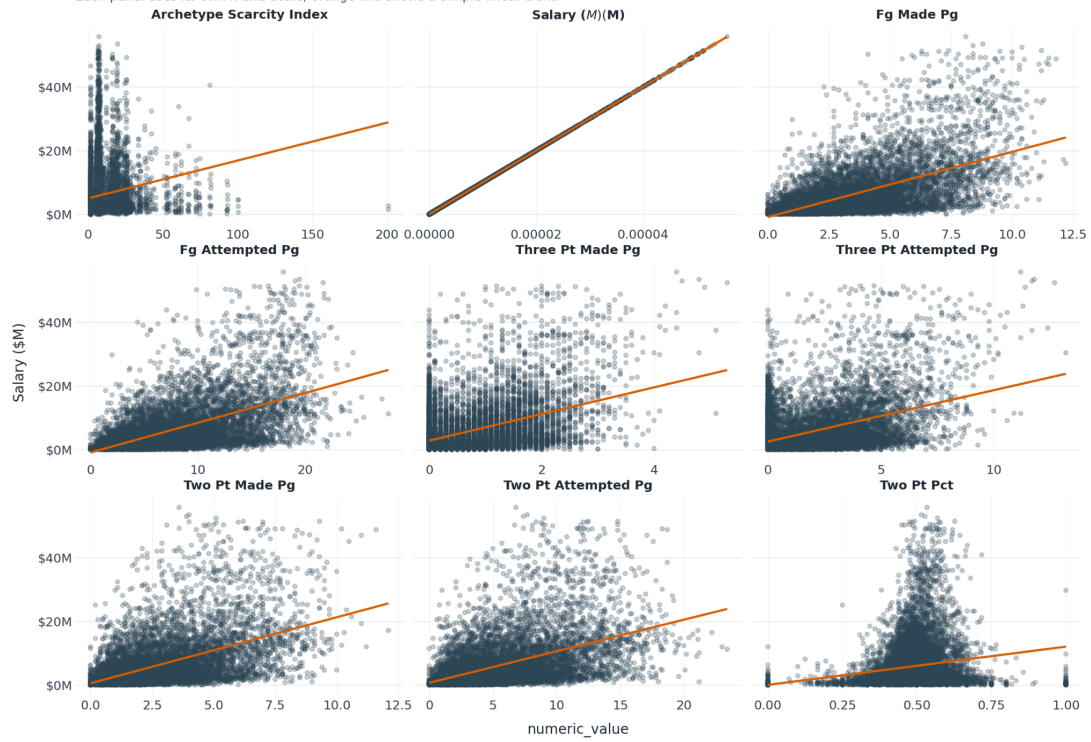
NBA Salary vs Numeric Variables (5 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



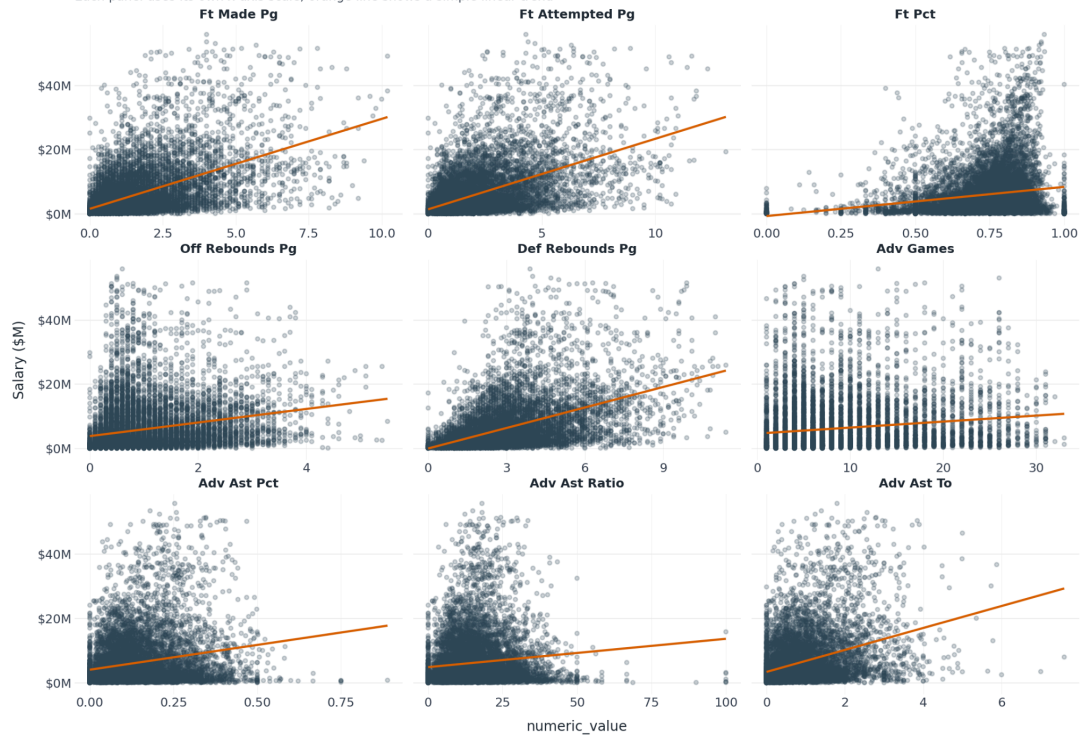
NBA Salary vs Numeric Variables (6 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



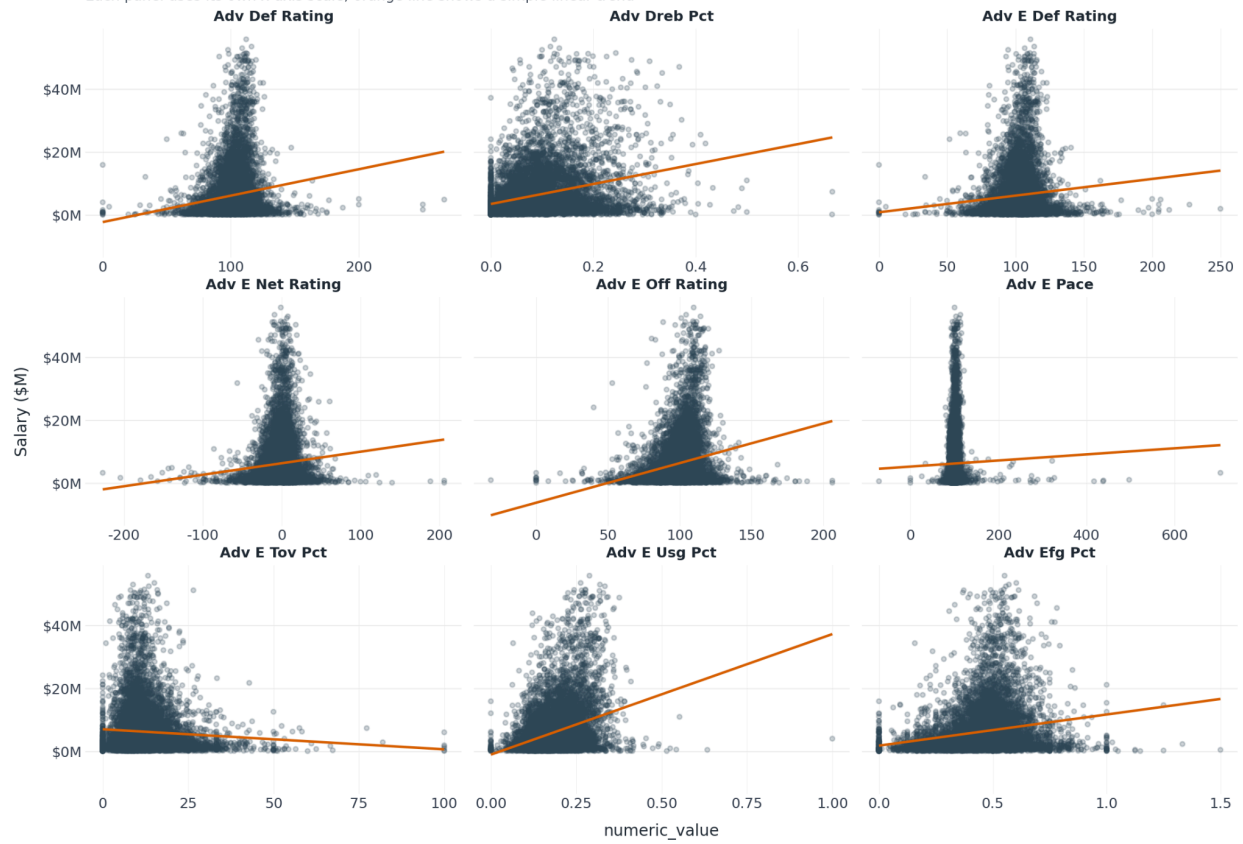
NBA Salary vs Numeric Variables (7 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



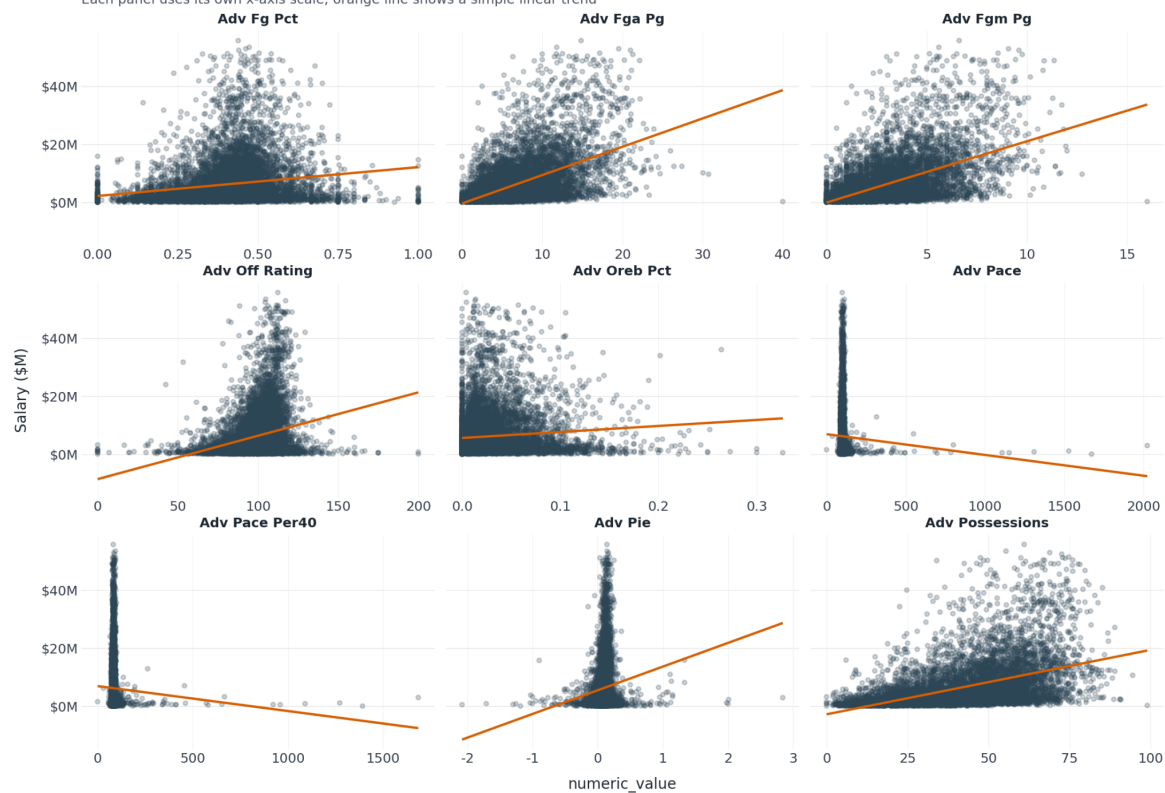
NBA Salary vs Numeric Variables (8 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



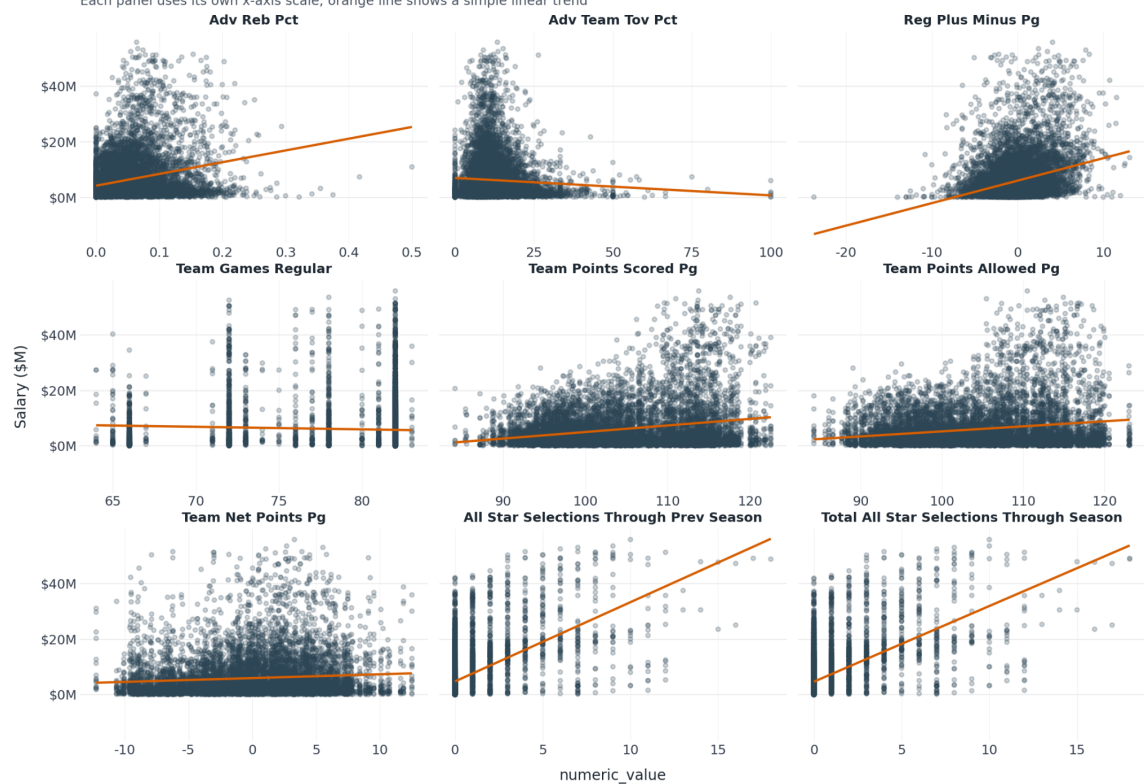
NBA Salary vs Numeric Variables (9 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



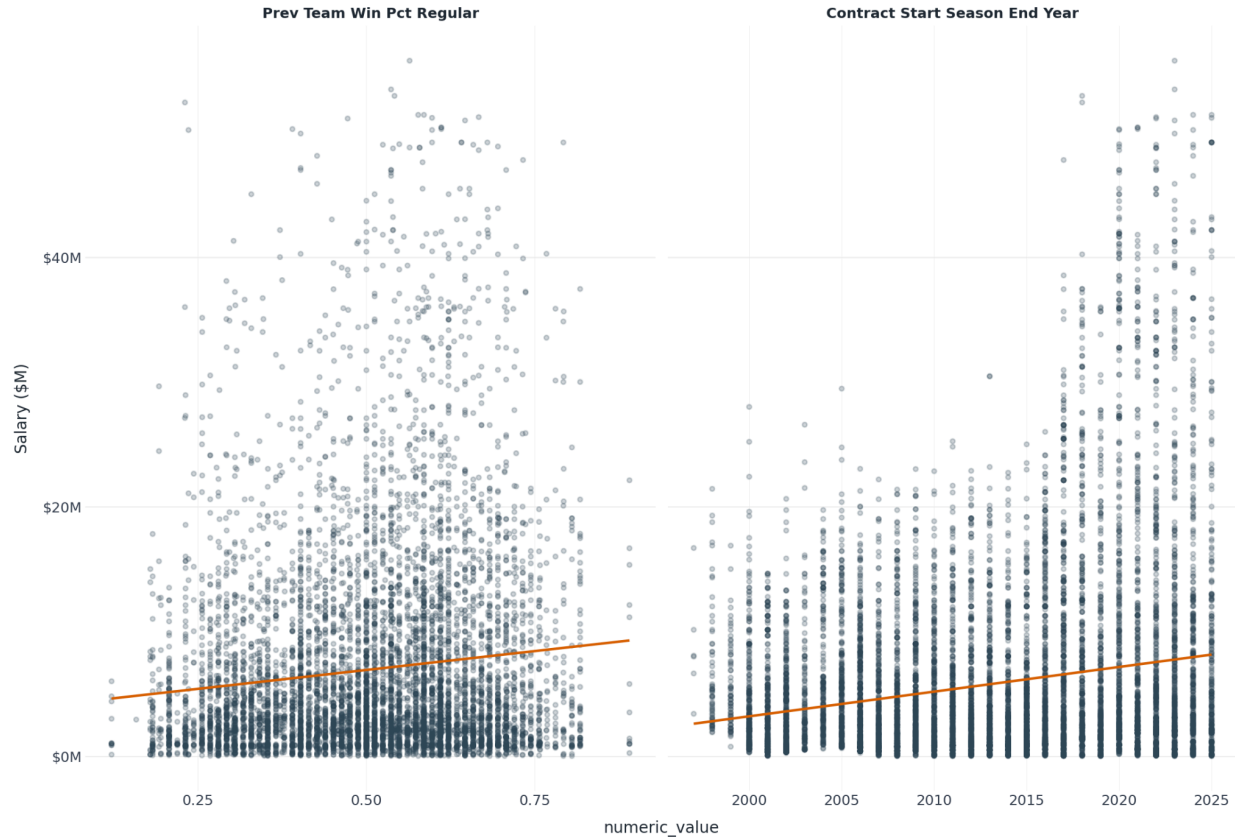
NBA Salary vs Numeric Variables (10 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



NBA Salary vs Numeric Variables (11 of 11)

Each panel uses its own x-axis scale; orange line shows a simple linear trend



Data Sources

1. NBA Player Stats and Salaries 2000-2025 (ratin21): This dataset served as the base dataset for the project because it already links NBA player salary information with season-level player performance data from 2000-2025. It provided the primary framework for connecting compensation outcomes with box-score and season summary statistics, making it the starting point for the final merged analysis dataset.
2. NBA Player Salaries (ulrikthgyepedersen): This source was used as a salary-focused reference dataset to strengthen salary coverage across seasons and to cross-check player values. It helped improved the accuracy and completeness of the target salary variable used in our predictive models.
3. NBA Players Salaries (omarsobhy14): This dataset was used as an additional salary and player-attribute source to improve metadata coverage and help validate salary records. It provided supporting information for resolving inconsistencies across salary-related fields and player identifiers.

4. NBA Dataset 1947-2025 (dahan1): This source was used for long-range season-level player statistics, awards, and career-history features. It allowed us to engineer long-term trend variables and reputation-based predictors, such as prior honors, cumulative career context, and historical performance indicators.
5. Historical NBA Data and Player Box Scores (eoimamoore): This dataset provided detailed game-level and box-score context that supported the creation of richer rolling and advanced performance features. It was especially useful for constructing trend-based measures and more detailed indicators of player contribution over time.
6. Basketball-Reference Advanced Player Stats (Season Tables): Basketball-Reference season-level advanced player statistics were used to add advanced efficiency and impact metrics to the merged dataset. These values supported the construction of Basketball-Reference-based features such as PER, BPM, VORP, and Win Shares-based variables that strengthened the performance driven modeling approach.
7. Basketball-Reference Salary Cap History: This source provided historical NBA salary cap values by season. These values were used to create season level financial context variables and may also support additional analysis related to league-wide salary environment and contract valuation.
8. Spotrac NBA CBA (Cap Rules Reference): Spotrac's CBA reference materials were used to determine player service-time groupings and CBA membership buckets. These buckets were useful for creating experience-related features that reflect how salary structures differ across rookie-scale, mid-career, and veteran player groups.
9. Spotrac NBA Luxury Tax History: This source provided historical luxury tax thresholds by season. These data added financial context at the team and league level and may help explain contract behavior under changing payroll constraints.
10. Basketball-Reference Player Contracts Index: This source was used as a manual reference for checking NBA player contracts across years. While it was not directly implemented into the final cleaned datasets, it served as a validation source during data review and reconciliation.
11. NBA Merged Data Source: The final dataset used for modeling was a cleaned and merged CSV compiled from the sources above. Useful variables from each source were integrated into this file, including salary data, player metadata, season-level statistics, advanced metrics, awards/history features, service-time indicators, and financial context variables. This merged dataset served as the direct input for our machine learning analysis and salary prediction models.

Proposed Modeling Approaches

XGBoost

XGBoost serves as a natural starting point for this project because it aligns closely with the complexity of how NBA salaries are actually determined. Rather than relying on simple linear relationships, salaries emerge from a mix of interacting factors, performance, role, durability, experience, and broader market dynamics. XGBoost is designed to uncover exactly these kinds of nonlinear patterns by building trees sequentially, where each new tree learns from the mistakes of the previous ones. This iterative refinement makes it especially powerful for a dataset that blends traditional stats with more nuanced contextual features. At the same time, XGBoost offers a level of control that is important for this project: its regularization tools help guard against overfitting, which is a real concern when working with many engineered variables or smaller positional subsets. Beyond prediction, it also gives us a window into interpretation, allowing us to identify which features matter most in shaping salary outcomes. That combination of predictive strength and interpretability makes it a strong anchor model, even though it requires careful tuning and can become computationally heavy as the modeling pipeline grows.

CatBoost

CatBoost brings a slightly different perspective to the problem, one that is particularly well-suited for messy, real-world data. NBA salary data is not only complex but also noisy, with inconsistencies across seasons, evolving roles, and uneven feature availability. CatBoost is designed with these challenges in mind, using an ordered boosting process that reduces the risk of information leakage and improves stability when making forward-looking predictions. This is especially relevant in a time-based setting like ours, where we are predicting future salaries from past performance. Another appealing aspect of CatBoost is its ability to handle categorical variables more naturally, which could become important if features like position, team, or contract context are incorporated more deeply. Even when the dataset leans heavily numerical, CatBoost tends to perform reliably with relatively minimal preprocessing, making it a useful complement to more tuning-intensive models. Its main drawbacks are practical: it can be slower to train at scale and slightly harder to interpret. Still, its robustness and stability make it a valuable model to include, particularly as a point of comparison.

Blended Model (CatBoost & XGBoost)

Rather than relying on a single model to capture the full complexity of NBA salaries, a blended approach allows us to combine different modeling strengths into a more balanced prediction. XGBoost and CatBoost, while similar in structure, often learn slightly different patterns from the same data. By averaging their predictions with carefully chosen weights, we

can reduce the impact of any one model's biases or overfitting tendencies. This is especially useful in a setting like ours, where salary outcomes are highly variable and influenced by factors that are only partially observed. Blending can help smooth out extreme predictions, such as overestimating max contracts or underestimating role players, and improve performance on unseen data. It also provides a kind of insurance against model-specific weaknesses, leading to more stable results across different player groups or eras. The tradeoff, however, is that this added stability comes at the cost of simplicity. Blended models are harder to interpret and explain, and their effectiveness depends heavily on how well the weights are tuned. Still, if the goal is to maximize predictive reliability, blending is a compelling strategy.

LightGBM

LightGBM plays a slightly different role in the project, acting as both a competitive model and a practical tool for rapid experimentation. Because the dataset will likely include a large number of features, ranging from box score stats to advanced metrics and contextual indicators, modeling speed becomes an important consideration. LightGBM is optimized for efficiency, using a histogram-based approach and a leaf-wise tree growth strategy that allows it to train quickly while still capturing complex relationships. This makes it especially useful during the iterative phases of the project, where features are being added, removed, and refined. Its ability to handle large-scale data efficiently enables more thorough experimentation within a limited timeframe. However, this efficiency comes with some caution. The same flexibility that allows LightGBM to fit complex patterns can also lead it to overfit, particularly in smaller subsets like positional groupings, if its parameters are not carefully controlled. There is also a risk that its speed encourages shallow exploration of the model space, prioritizing quick results over deeper optimization. Even so, LightGBM remains an important part of the modeling toolkit, balancing performance with practicality in a data-rich environment.

Learning Goals

Sahil

Amrit

Diya

Dirks

One of my primary learning goals for this project is to deepen my understanding of feature engineering, particularly in the context of a complex, real-world dataset like NBA player salaries. The success of our models will depend heavily on how well we can construct meaningful features from the available data. This includes not only selecting relevant performance metrics, but also designing variables that better capture player role, consistency, durability, and contextual factors such as era or positional scarcity. I'm especially interested in experimenting with transformations like rolling averages, per-minute or per-possession statistics, interaction terms, and era-adjusted metrics to see how they influence model performance. Through this process, I hope to better understand how thoughtful feature construction can improve predictive power and better represent the underlying dynamics of player value.

I want to develop a clearer intuition for the strengths and tradeoffs of different machine learning models, particularly tree-based methods like XGBoost, CatBoost, and LightGBM. While all three are designed for tabular data, they differ in how they handle feature interactions, noise, categorical variables, and computational efficiency. By comparing their performance across different feature sets and validation schemes, I aim to learn when and why one model may outperform another, and how model choice interacts with the quality of the features themselves. I'm also interested in exploring how techniques like blending can combine these strengths to produce more stable predictions. Overall, this project is an opportunity to move beyond treating models as "black boxes" and instead develop a more nuanced understanding of how data preparation and model selection work together to drive results.